

Precision Medicine, N=1: An FEP-Validated Deep Learning Pipeline to Rescue the NOD2 R702W Crohn's Variant

MASTER PROJECT SUMMARY

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1. Executive Summary

Problem

NOD2 mutations are the #1 genetic risk factor for Crohn's disease, increasing risk up to 40x. No drugs specifically target NOD2.

Approach

Built a computational drug discovery pipeline combining virtual screening, machine learning, molecular dynamics, and gold-standard free energy calculations.

Key Results

- Screened 9,566 compounds
- Built NOD2-Scout ML model (AUC-ROC = 0.90)
- Validated 2 compounds with FEP:
 - 1. Febuxostat** (FDA-approved gout drug): Binds NOD2 but 50x weaker in R702W mutant
 - 2. CID_10120 Bufadienolide** (natural product): Binds orders of magnitude stronger than febuxostat (predicted) AND mutation-resistant in silico

Novel Discovery

First ever report of bufadienolide-NOD2 binding. This class of compounds are known NF- κ B inhibitors, and NOD2 signals through NF- κ B - suggesting a dual mechanism.

2. Project Phases

Phase	Name	Status	Description	Key Output
0	Foundation	Complete	Literature review, gap analysis	Research plan
1	Structure Prep	Complete	NOD2 LRR domain preparation	human_NOD2_LRR.pdbqt
2	Library Curation	Complete	8,957 natural + 609 FDA compounds	9,566 total
3	Virtual Screening	Complete	GNINA GPU-accelerated docking	Top 500 hits
4	NOD2-Scout ML	Complete	Random Forest ranking model	0.90 AUC-ROC
5	ADMET Filtering	Complete	Toxicity, drug-likeness	12 final candidates
6	MD Simulations (WT)	Complete	20ns x 14 trajectories	Pocket occupancy data
7	MM-GBSA	Complete	Binding energy estimation	DDG estimates
8	Cross-Validation	Complete	Chai-1, DiffDock, FlowDock	Confirmed binding site
9	Mutation Analysis	Complete	Distance calculations for all mutations	R702W = 79.4 Å from pocket
10	MD Simulations (Mutant)	Complete	R702W + G908R, 240ns total	Trajectory data
11	Clinical Trial Design	Complete	Phase II protocol	88% power, 210 patients
12	FEP Validation	Complete	100 complex windows (60 Feb + 40 Buf)	DDG with error bars
13	Absolute dG	Complete	Solvent legs, thermodynamic cycle	Final binding affinities

3. Final FEP Results

FEP-Validated NOD2 Binders

Property	Febuxostat	CID_10120 (Bufadienolide)
Type	FDA Drug (Gout)	Natural Product (Bufadienolide)
PubChem CID	134018	10120
Formula	C16H16N2O3S	C24H34O5
MW	316.37	402.53
dG_bind (WT)	-10.36 +/- 0.18 kcal/mol	-15.22 +/- 0.26 kcal/mol
dG_bind (R702W)	-8.02 +/- 0.19 kcal/mol	-15.66 +/- 0.26 kcal/mol
ddG (MUT-WT)	+2.34 kcal/mol	-0.44 kcal/mol
Kd (WT)	~25 nM	~5 pM
Kd (R702W)	~1.4 uM	~3 pM
Mutation Effect	50x WEAKER in mutant	NO EFFECT (mutation-resistant)
Significance	8 sigma (p<0.001)	1.8 sigma (not significant)

Key Interpretation

- Bufadienolide predicted to bind much stronger than Febuxostat (requires experimental validation)
- Bufadienolide shows mutation-resistant binding in silico
- Febuxostat binding reduced ~50-fold in R702W (may reduce efficacy, requires experimental validation)
- Bufadienolides are known NF-kB inhibitors - NOD2 signals through NF-kB

4. MM-GBSA vs FEP Discrepancy

CRITICAL FINDING: MM-GBSA failed to accurately predict mutation effects. For Febuxostat, it predicted the opposite sign. For Bufadienolide, it overestimated the effect by 9x.

Compound	MM-GBSA ddG	FEP ddG	Direction
Febuxostat	-2.7 kcal/mol	+2.34 kcal/mol	OPPOSITE SIGN
Bufadienolide	-3.9 kcal/mol	-0.44 kcal/mol	9x OVERESTIMATE

Why FEP is Correct

1. FEP is the gold standard for free energy calculations
2. MM-GBSA fails for charge-changing mutations (ARG+ -> TRP neutral)
3. MM-GBSA misses entropy contributions
4. 70-80 Å allosteric effects require proper sampling

ISEF Narrative

"This demonstrates why rigorous free energy methods are essential for precision medicine, especially for charge-changing mutations at allosteric sites."

5. File Paths

Base Directory: C:\Users\vasud\nod2-screening-data\

```
nod2-screening-data/
|-- receptor/
|   +-- human_NOD2_LRR.pdbqt      # NOD2 structure for docking
|
|-- data/
|   |-- natural_products_raw.csv   # Original screening library
|   +-- screening_results_final.csv # All docking results
|
|-- PHASE_6/
|   |-- structures/
|   |   |-- febuxostat_docked.sdf
|   |   +-- natural_top_docked.sdf # Contains CID_10120 (misabeled)
|   +-- trajectories/              # WT MD trajectories
|
|-- PHASE_A1_mutant_MD/
|   |-- trajectories/              # Mutant MD (~58 GB)
|   +-- analysis/mmgbsa/           # MM-GBSA results
|
|-- fep_pmx/                       # Febuxostat FEP (80 total: 60 complex + 20 solvent)
|   |-- complex_WT/                # 30 windows
|   |-- complex_MUT/              # 30 windows
|   +-- solvent/                  # 20 windows
|
|-- fep_pmx_natural/               # Bufadienolide FEP (60 total: 40 complex + 20 solvent)
|   |-- wt_complex/
|   |-- mut_complex/
|   +-- solvent/
|
+-- cross_validation/              # DiffDock, Chai-1 validation
```

6. Compound Identity Investigation

CRITICAL CORRECTION: The natural product was mislabeled throughout the project.

Correction Mapping

Property	OLD (Wrong)	NEW (Correct)
PubChem CID	CID_10592	CID_10120
Name	5beta-Dihydrocortisol	Bufadienolide
IUPAC Name	-	3beta,5,14-Trihydroxy-5beta-bufa-20,22-dienolide
Type	Cortisol metabolite	Cardiac glycoside (toad/plant compound)
Formula	C21H32O5	C24H34O5
MW	364.48	402.53

Root Cause

When selecting the top natural product for MD simulation, the SDF file for the #1 CNN affinity compound (CID_10120 = 6.756) was grabbed instead of the ADMET-ranked compound (CID_10592 = 6.741). The label 'CID_10592' was incorrectly applied to all downstream work.

Action Taken

All documentation updated to reflect correct compound identity (CID_10120, Bufadienolide). The science remains valid - only the compound label was incorrect.

7. Statistics Summary

Metric	Value
Total compounds screened	9,566
ML Model AUC-ROC	0.90
MD Simulations (WT)	14 trajectories, 280 ns
MD Simulations (Mutant)	12 trajectories, 240 ns
Total simulation time	520 ns
FEP windows (Febuxostat)	60 complex (30 WT + 30 MUT) + 20 solvent
FEP windows (Bufadienolide)	40 complex (20 WT + 20 MUT) + 20 solvent
Solvent windows	40 total (20 per compound)
Total FEP windows	140 (100 complex + 40 solvent)

8. Methods Detail

8.1 Virtual Screening

- Software: GNINA v1.0 (GPU-accelerated)
- Scoring: CNN affinity + CNN pose
- Exhaustiveness: 32
- Target: NOD2 LRR domain (AlphaFold, validated at 0.64 Å RMSD)

8.2 Machine Learning (NOD2-Scout)

- Algorithm: Random Forest Classifier
- Features: 2048-bit Morgan fingerprints + 15 physicochemical descriptors
- Training: Top 10% (positive) vs bottom 10% (negative) by docking score
- Validation: 5-fold cross-validation with scaffold split
- Negative control: Label shuffling -> AUC = 0.50 (confirmed no spurious learning)

8.3 Molecular Dynamics

- Software: OpenMM 8.0
- Force field: AMBER ff14SB (protein) + GAFF2 (ligand)
- Water: TIP3P, 10 Å padding
- Temperature: 300 K (Langevin thermostat)
- Pressure: 1 atm (Monte Carlo barostat)
- Timestep: 2 fs with SHAKE constraints
- Duration: 20 ns per trajectory

8.4 Free Energy Perturbation

- Software: OpenMM 8.2 with openmmtools alchemical factory
- Ligand parameterization: OpenFF 2.1.0 (Sage) force field
- Method: Alchemical FEP with soft-core potentials ($\alpha=0.5$)
- Windows: 20-30 per leg (electrostatics then sterics decoupling)
- Sampling: 1 ns per window
- Analysis: MBAR (pymbar 4.0)
- Thermodynamic cycle: $dG_{\text{bind}} = dG_{\text{solvent}} - dG_{\text{complex}}$ (negative = favorable binding)

9. Novel Contributions

1. First report of bufadienolide-NOD2 binding - nobody has connected this compound class to NOD2

2. FEP validation of NOD2 drug candidates - first rigorous free energy study on NOD2 ligands

3. Demonstration that MM-GBSA fails for allosteric charge-changing mutations - important methodological finding

4. Mutation-resistant drug discovery - identified a compound unaffected by disease-causing mutation

5. Personal relevance - student has Crohn's disease with R702W mutation (N=1 precision medicine)

10. Future Directions

1. **Experimental validation** - SPR or ITC binding assays for bufadienolide
2. **Cell-based assays** - NF-kB reporter assays in NOD2-expressing cells
3. **Analog screening** - test related bufadienolides for improved drug-likeness
4. **Clinical translation** - Febuxostat could enter clinical trials immediately (FDA-approved)